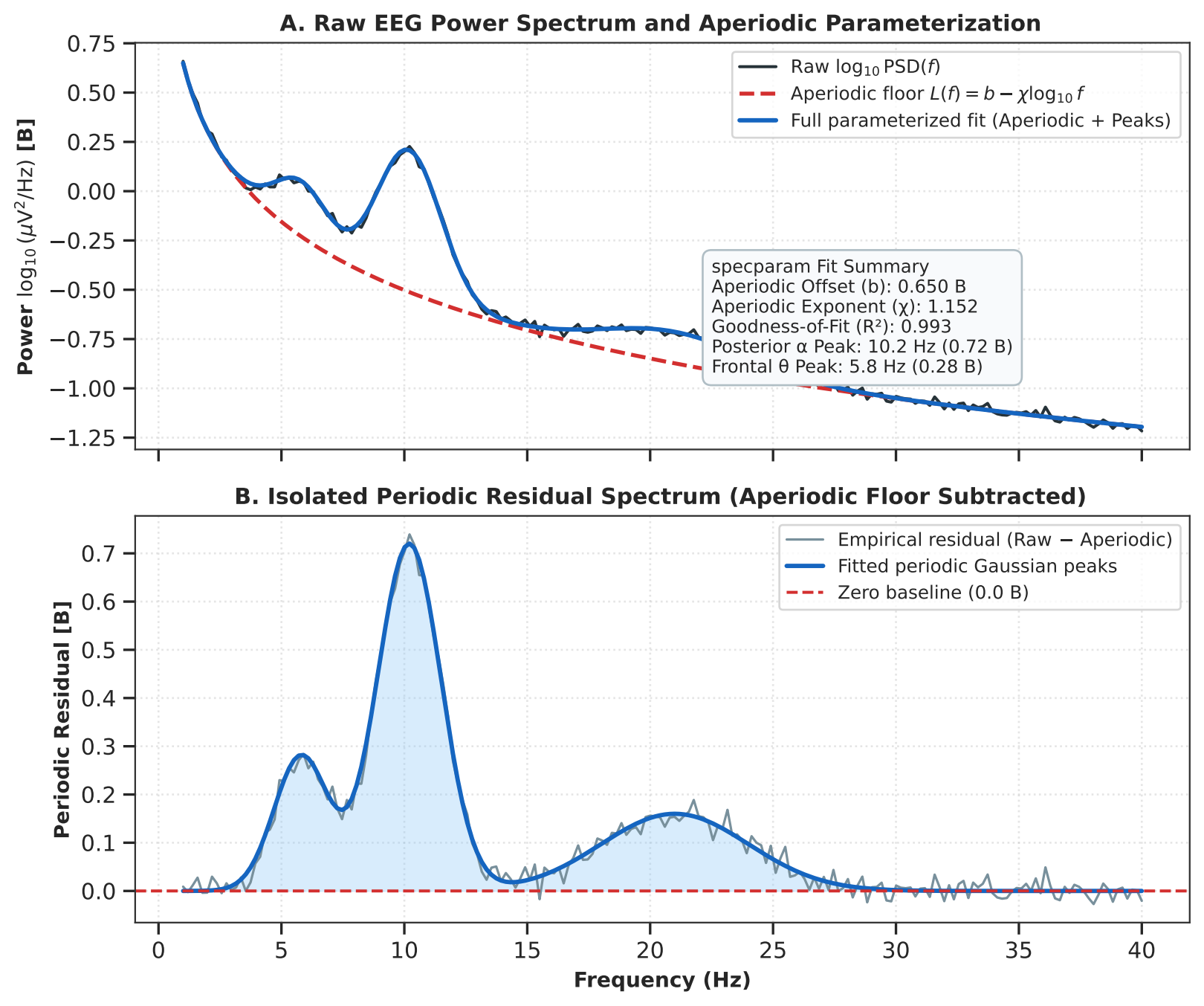
A. Raw EEG Power Spectrum and Aperiodic Parameterization
Power $\log_{10}(\mu V^2/Hz)$ [B]
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
Raw $\log_{10}$ PSD(f)
Aperiodic floor $L(f) = b - \chi \log_{10} f$
Full parameterized fit (Aperiodic + Peaks)
specparam Fit Summary
Aperiodic Offset (b): 0.650 B
Aperiodic Exponent (χ): 1.152
Goodness-of-Fit (R^2): 0.993
Posterior α Peak: 10.2 Hz (0.72 B)
Frontal θ Peak: 5.8 Hz (0.28 B)
B. Isolated Periodic Residual Spectrum (Aperiodic Floor Subtracted)
Periodic Residual [B]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Empirical residual (Raw − Aperiodic)
Fitted periodic Gaussian peaks
Zero baseline (0.0 B)
0
5
10
15
20
25
30
35
40
Frequency (Hz)